

Three compositional, tree-aware VAEs — at a glance

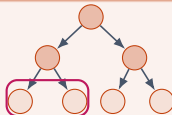
PhILR-VAE



Make the geometry flat, then use a vanilla VAE.

- **Tree** → an orthonormal *balance basis* Ψ .
- **Recon**: logistic-normal — exact, no extra penalty.

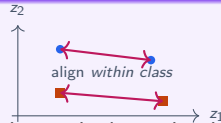
TreeDTM-VAE



Put a likelihood at every node of the tree.

- **Tree** → one local *split* per internal node.
- **Recon**: per-clade Dirichlet-tree-multinomial.

CAPDA-VAE



Erase the study, keep the disease signal.

- **Input**: multi-resolution taxonomy (clr + aggregates).
- **Align** same-class latents *across studies* (no adversary).

Shared backbone: simplex data (*only ratios inform*) · Gaussian latent $\mathcal{N}(\mathbf{0}, I)$ · ELBO with β warm-up.